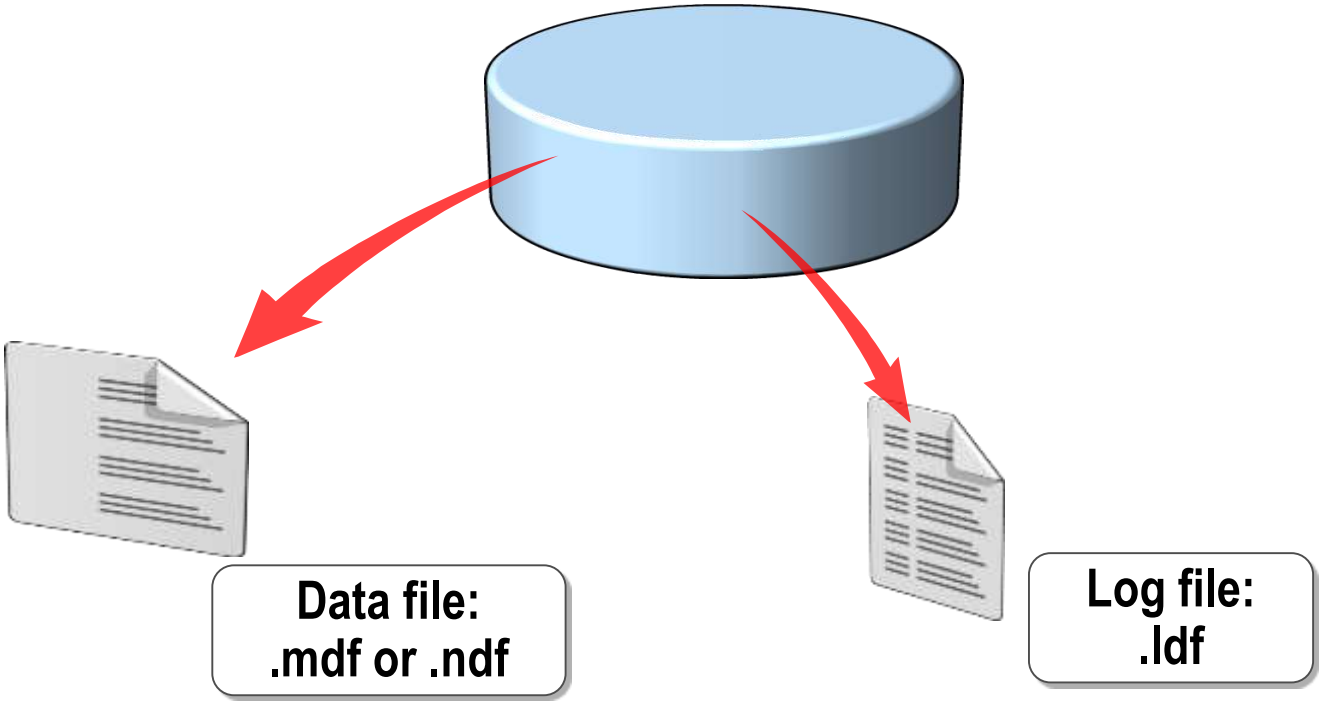
MANAGING DATABASES, DATABASE FILES AND TABLES

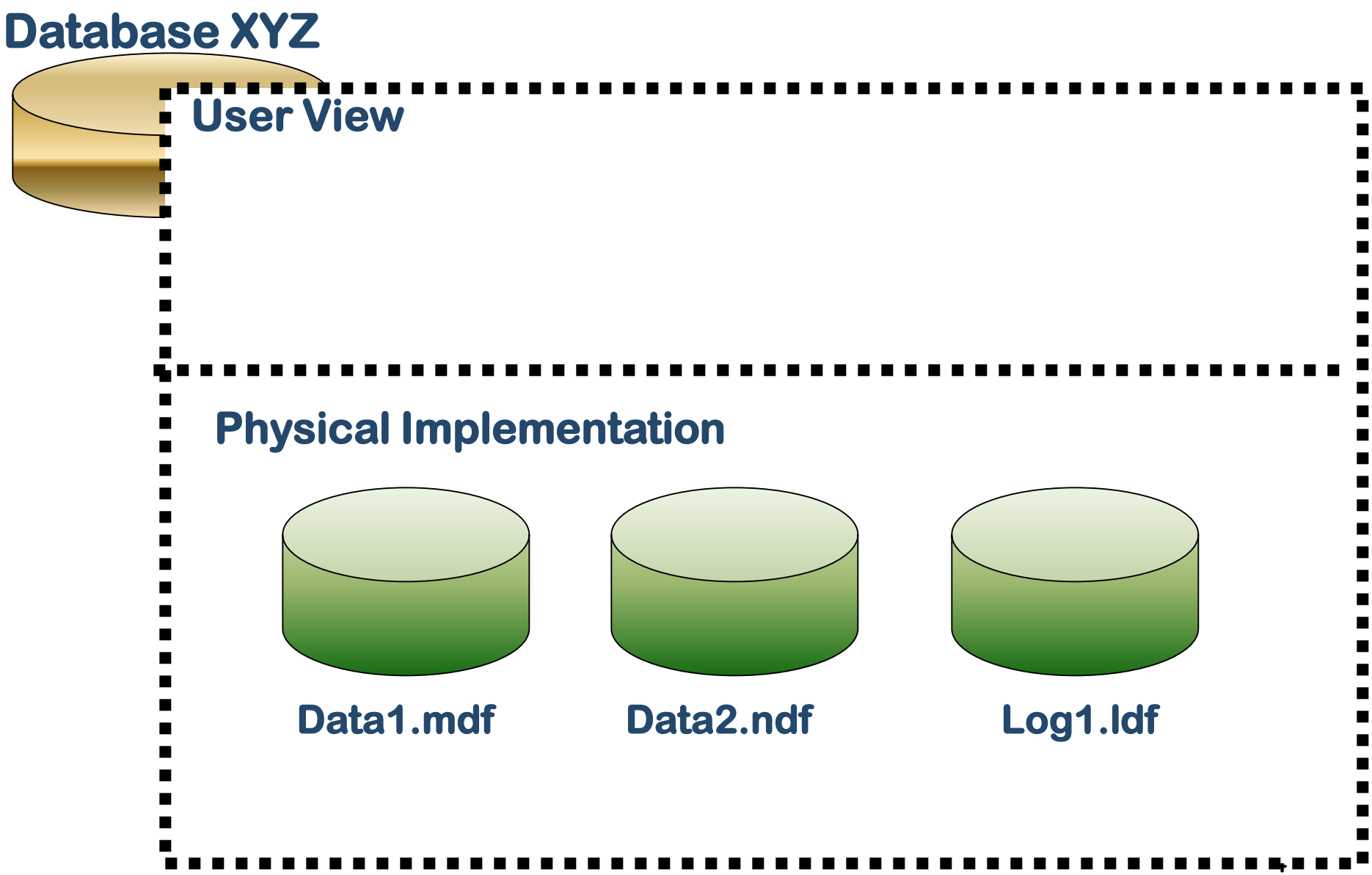
Objectives

- **Physical Database Files**
- **Creating Databases**
- **Managing Databases**
- **Managing Tables**

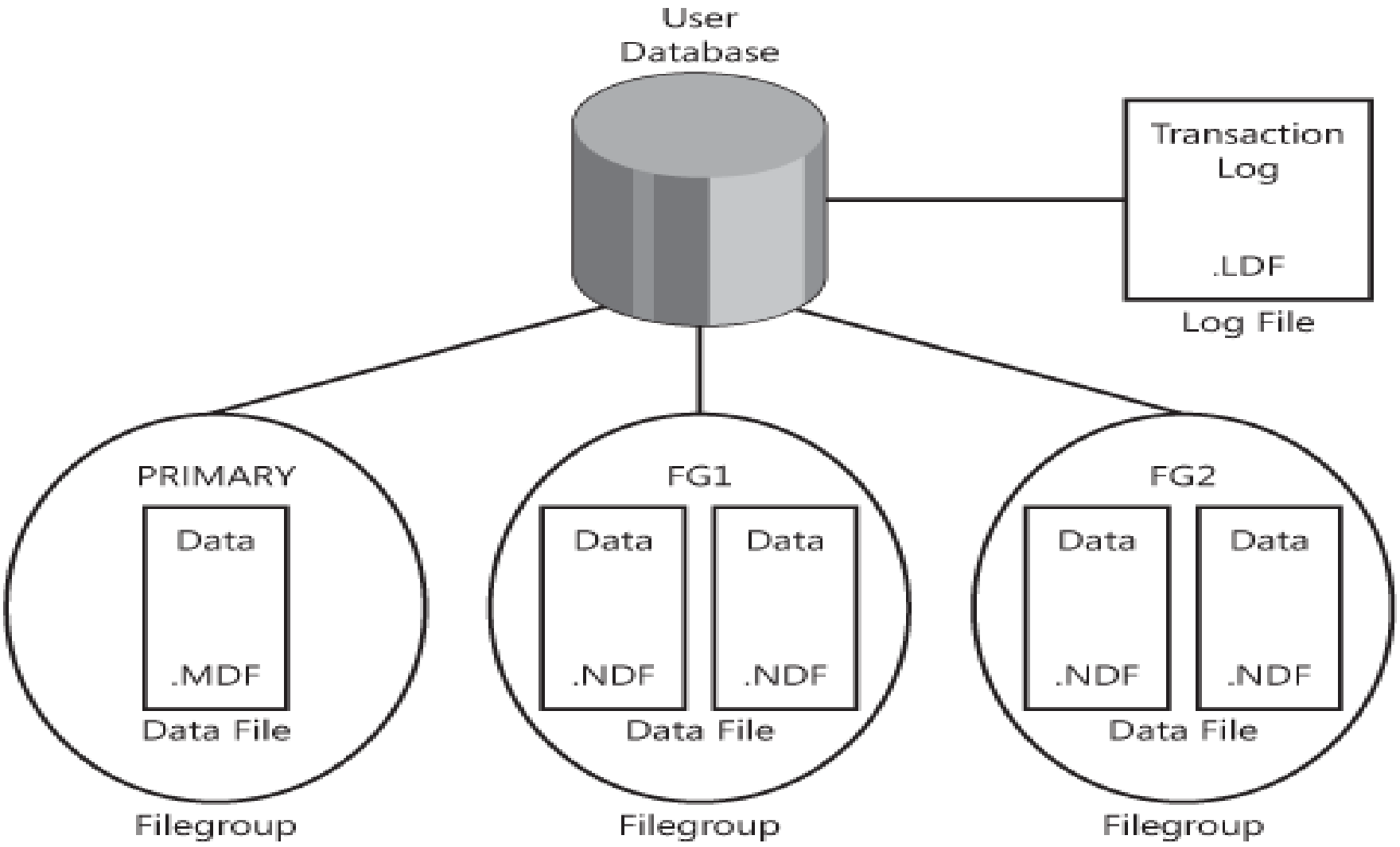
Database



Database


Database XYZ
User View
Physical Implementation
Data1.mdf
Data2.ndf
Log1.ldf

Database Layout



Physical Database Files

Microsoft® SQL Server™ 2012 maps a database over a set of operating-system files. Data and log information are never mixed on the same file, and individual files are used only by one database.

Types of Physical Database Files

Primary data files

- **The primary data file is the starting point of the database and points to the other files in the database.**

- **Every database has one primary data file.**

- **The recommended file name extension for primary data files is .mdf.**

Types of Physical Database Files

Secondary data files

- **comprise all of the data files other than the primary data file. Some databases may not have any secondary data files, while others have multiple secondary data files.**

- **The recommended file name extension for secondary data files is .ndf.**

Types of Physical Database Files

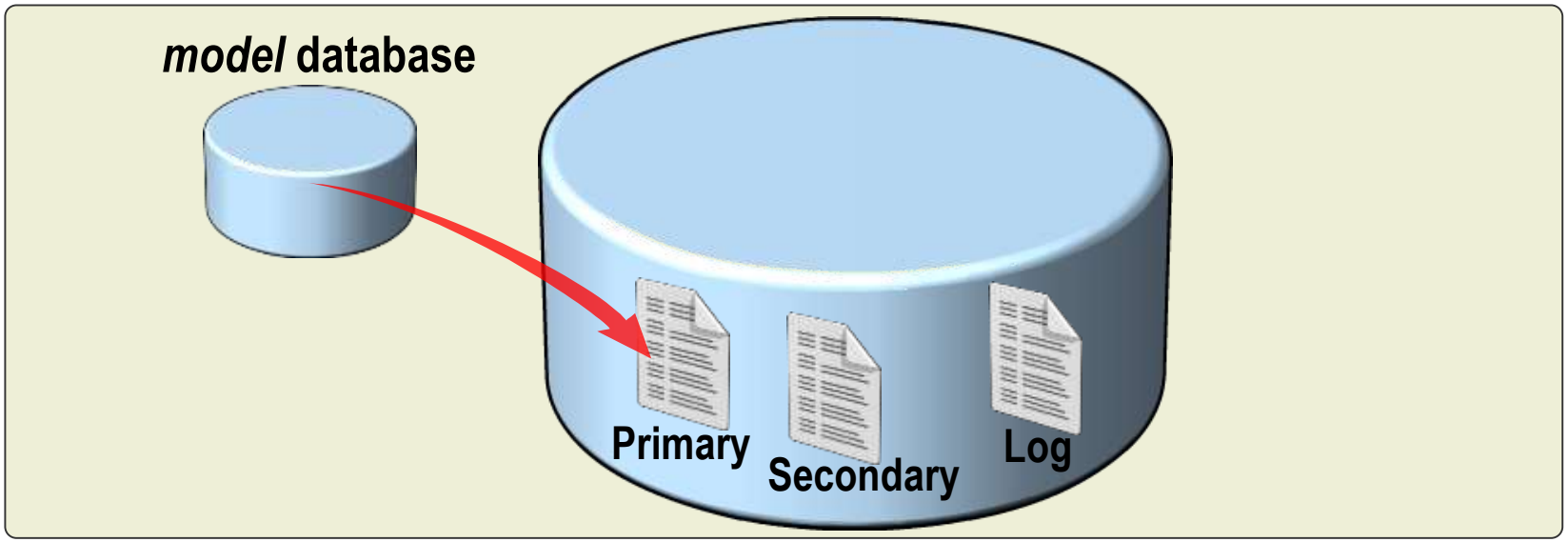
Log files

- **used to store changes to the database before these changes are recorded in the data files themselves**

- **It enables SQL Server to use these files as an important part of its recovery process**

- **The recommended file name extension for log files is .ldf.**

Considerations for Creating a Database



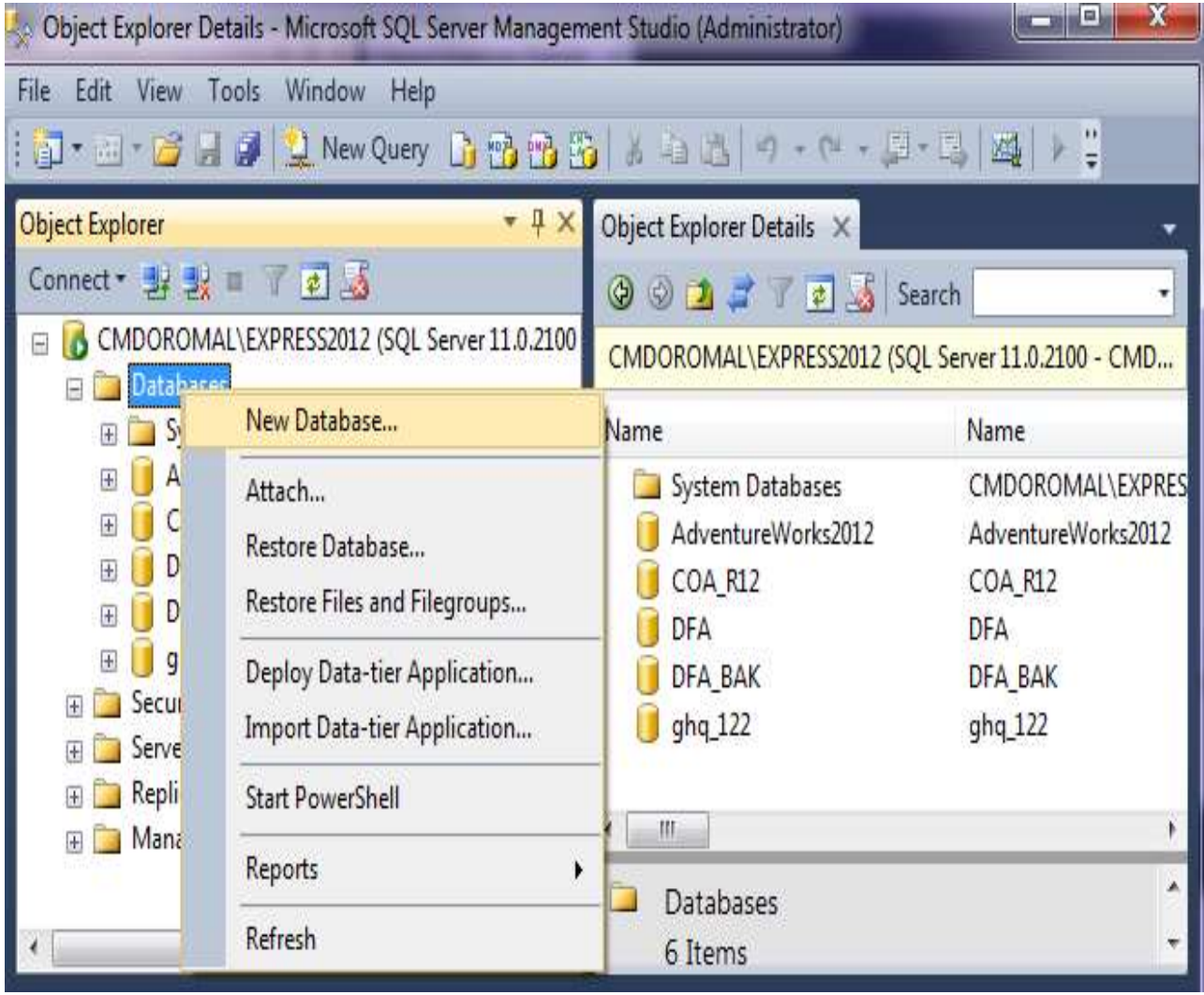
- **Database files include primary, secondary, and log files**
- **Considerations for database size**
  - **Initial size of data files**
  - **Initial size of log files**
  - **Potential growth of physical data storage**

Creating a Database

Using Object Explorer

1. **Right-click either the Databases folder in the console tree or the white space in the right pane**

2. **Choose New Database…**

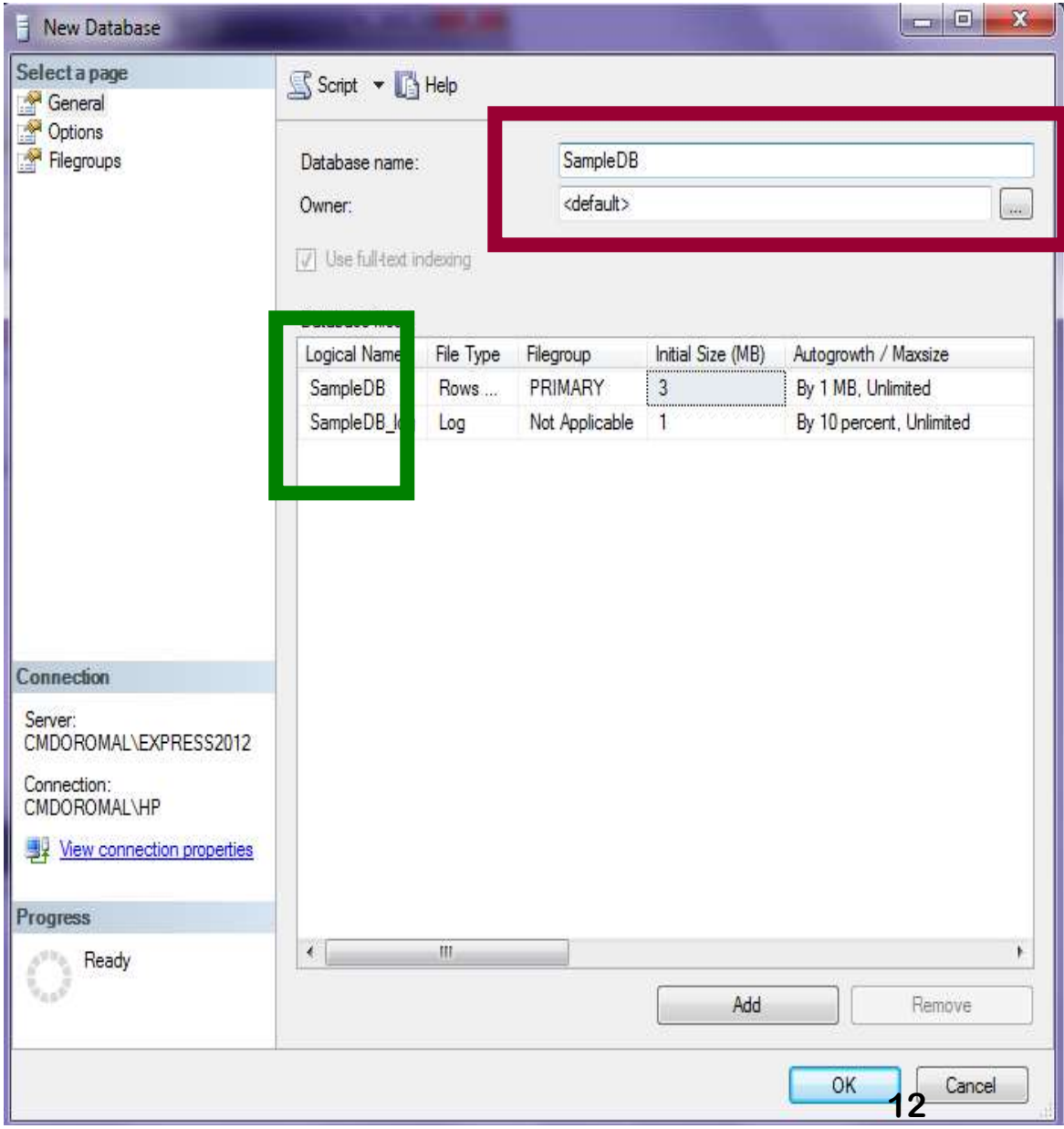


Creating a Database

Using Object Explorer

3. Enter a name for the database and leave the owner as <default>.

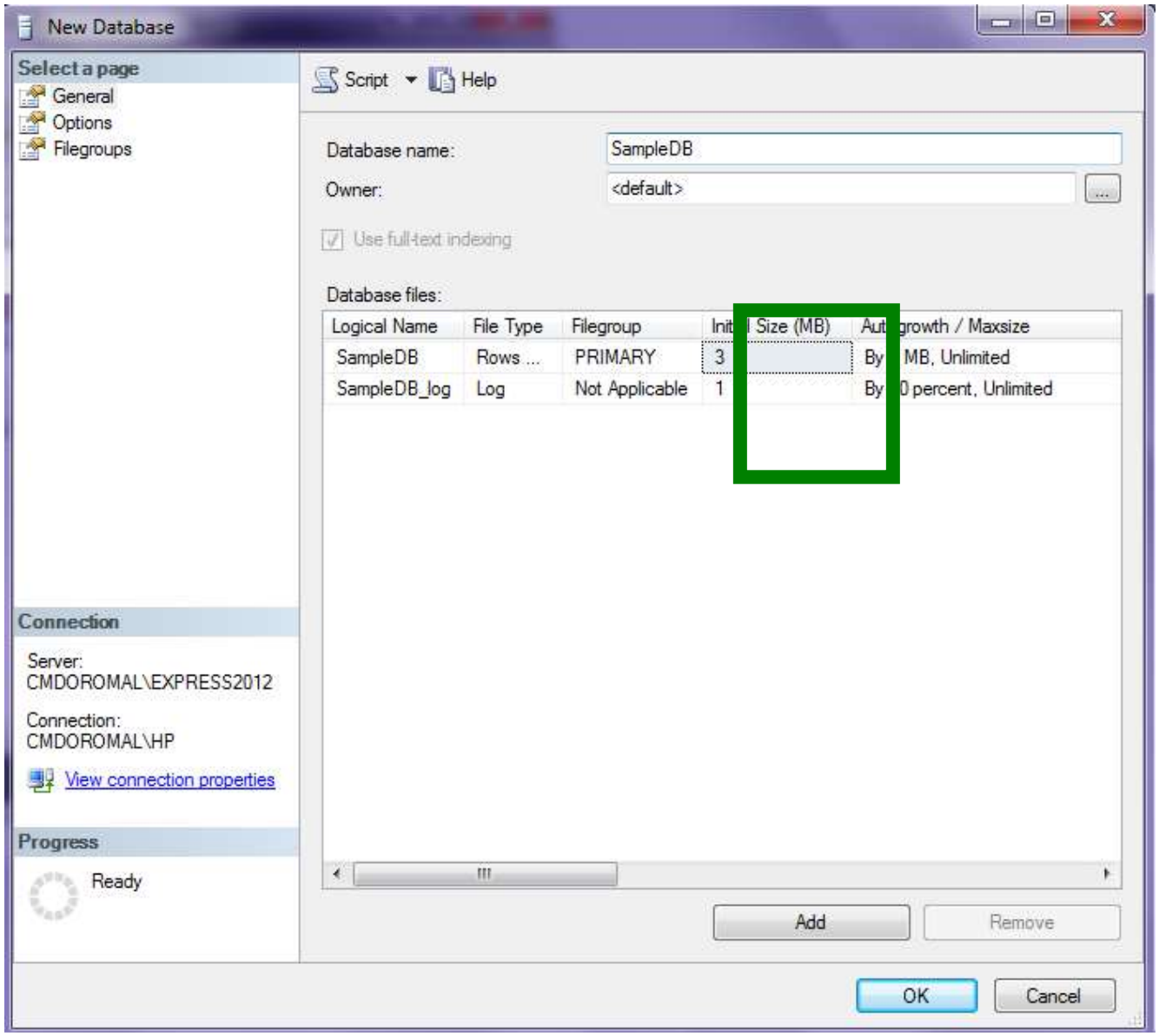
4. On the Database files, Logical Name automatically created based on your database name



Creating a Database

Using Object Explorer

5. On the Initial Size column, input the size of the file in MB

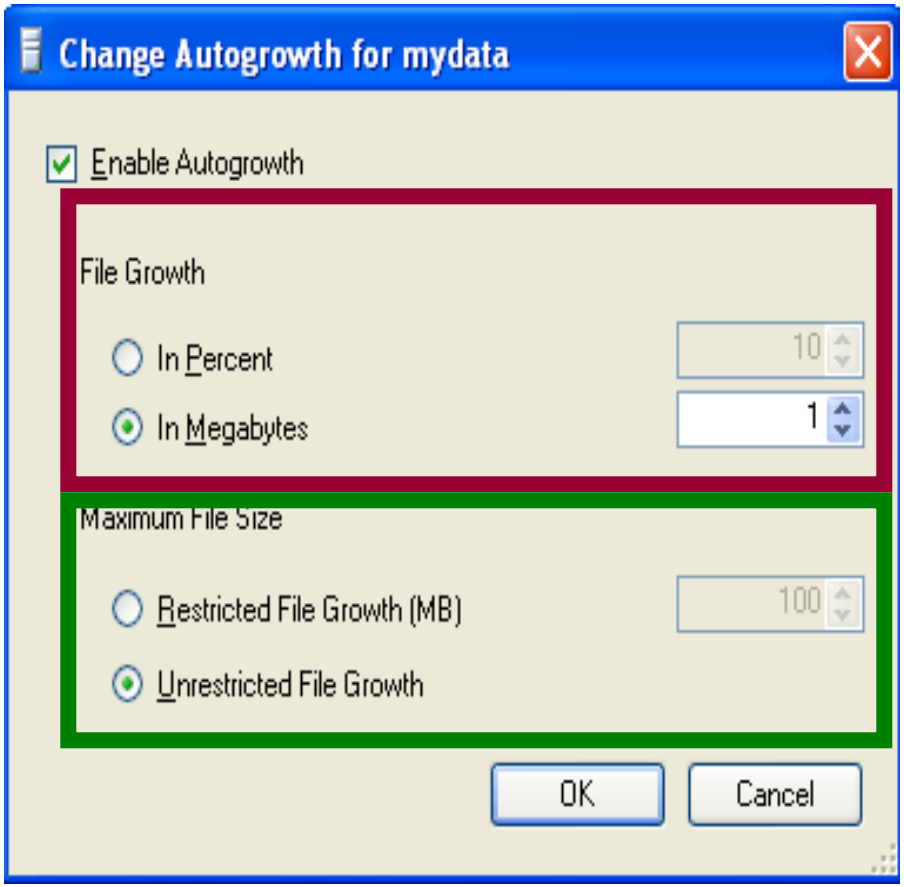


Creating a Database

Using Object Explorer

6. In the Autogrowth column, click the ellipsis button. A dialog box pops up, choose **File Growth** in Megabytes or by Percent

7. Select **Maximum File Size** whether Unrestricted file growth or Restricted file growth (MB)



Creating a Database

Query Editor (T-SQL)

```
CREATE DATABASE database_name
  [ON ...n ]
   [PRIMARY ]
   ([ NAME = logical_file_name,]
     FILENAME = 'physical_file_name'
     [, SIZE = size ]
     [, MAXSIZE = maxsize ]
     [, FILEGROWTH = growth_increment ]
```

Creating a Database

database_name

- **name of the new database.**

PRIMARY

- **Specifies that associated file list defines the primary file.**

NAME

- **Specifies the logical name for the file**

logical_file_name

- **Is the name used to reference the file any T-SQL executed after the DB is created**

Creating a Database

FILENAME

- **Specifies the physical name for the file**

physical_file_name

- **Is the path and file name of the physical file**

SIZE

- **Specifies the size the file**

size

- **Initial size of the file. Default value 1MB**

Creating a Database

MAXSIZE

- Specifies the maximum size of the file

max_size

- Initial maximum size of the file. If not specified, the file grows until the disk is full

FILEGROWTH

- Specifies the growth increment of the file

growth_increment

- The amount of space added to the file each time new space is needed. Minimum value is 64 KB. Default is 10% of the file

Creating a Database

Query Editor (T-SQL)

```
CREATE DATABASE MyDatabase
ON
 PRIMARY (NAME = 'DataStore',
    FILENAME = 'd:\data
       directory\DataStore_MyDatabase.mdf',
    SIZE = 2MB, MAXSIZE = 5MB,
    FILEGROWTH = 1MB)
LOG ON
     (NAME ='LogStore',
       FILENAME = 'e:\log
         directory\LogStore_MyDatabase.ldf',
       SIZE = 1MB, MAXSIZE = 5MB,
       FILEGROWTH = 1MB)
```

Viewing Information of DB

Query Editor (T-SQL)

```
sp_helpdb database_name
```

- **The sp_helpdb stored procedure used by itself will give you information about all databases in your SQL Server**
- **You can gather information about a particular database by using the database name as a parameter**

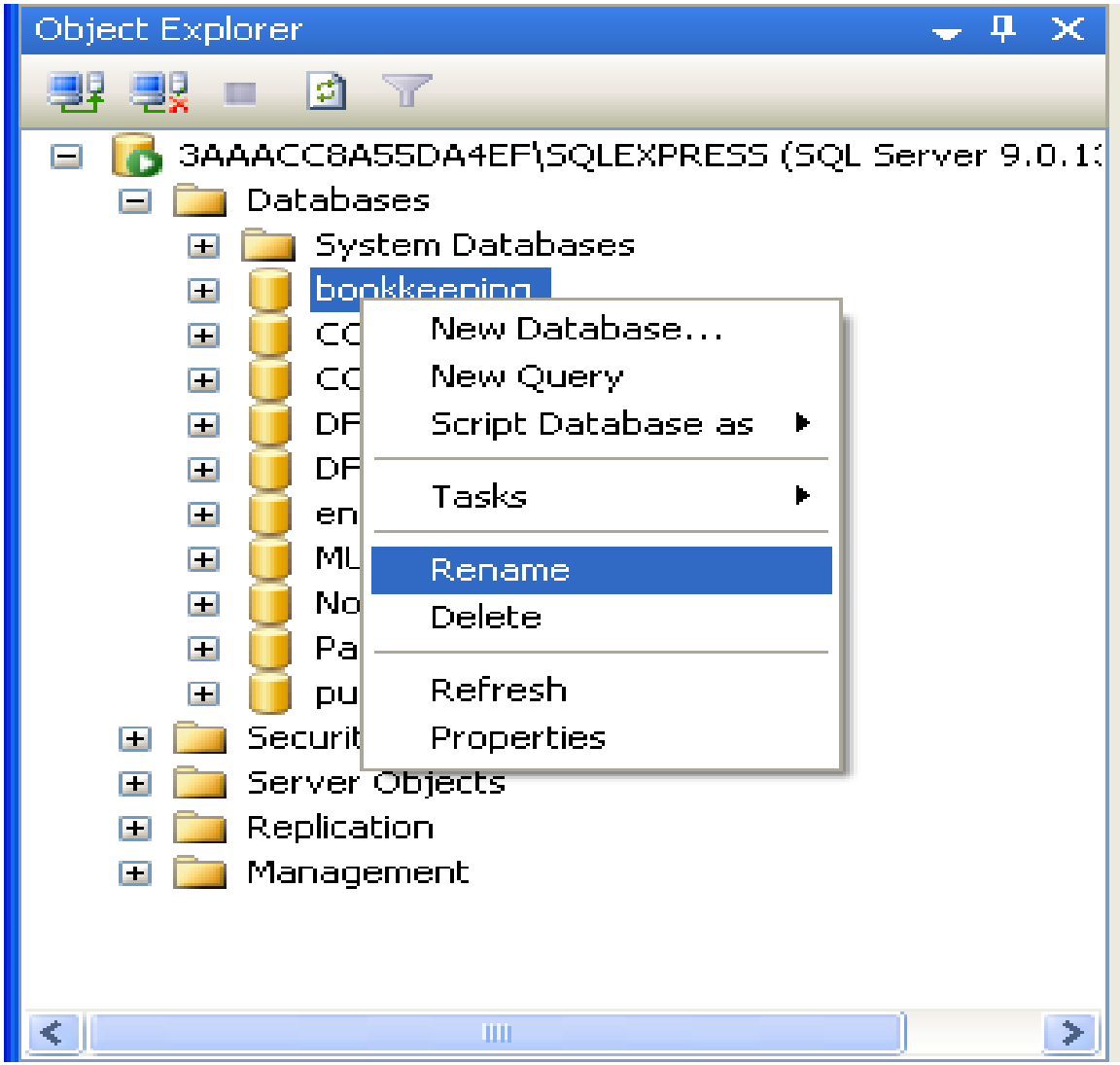
Renaming Database

Using Object Explorer

1. **Right-click on the databases to be deleted**

2. **Choose Rename**

3. **Input the new name of the database**



Renaming Database

Query Editor (T-SQL)

sp_renamedb

- **change the name of the database**

Syntax:

```
sp_renamedb [ @dbname = ] 'old_name' ,
   [ @newname = ] 'new_name'
```

Renaming Database

Query Editor (T-SQL)

Change the name of the *accounting* database to *financial*.

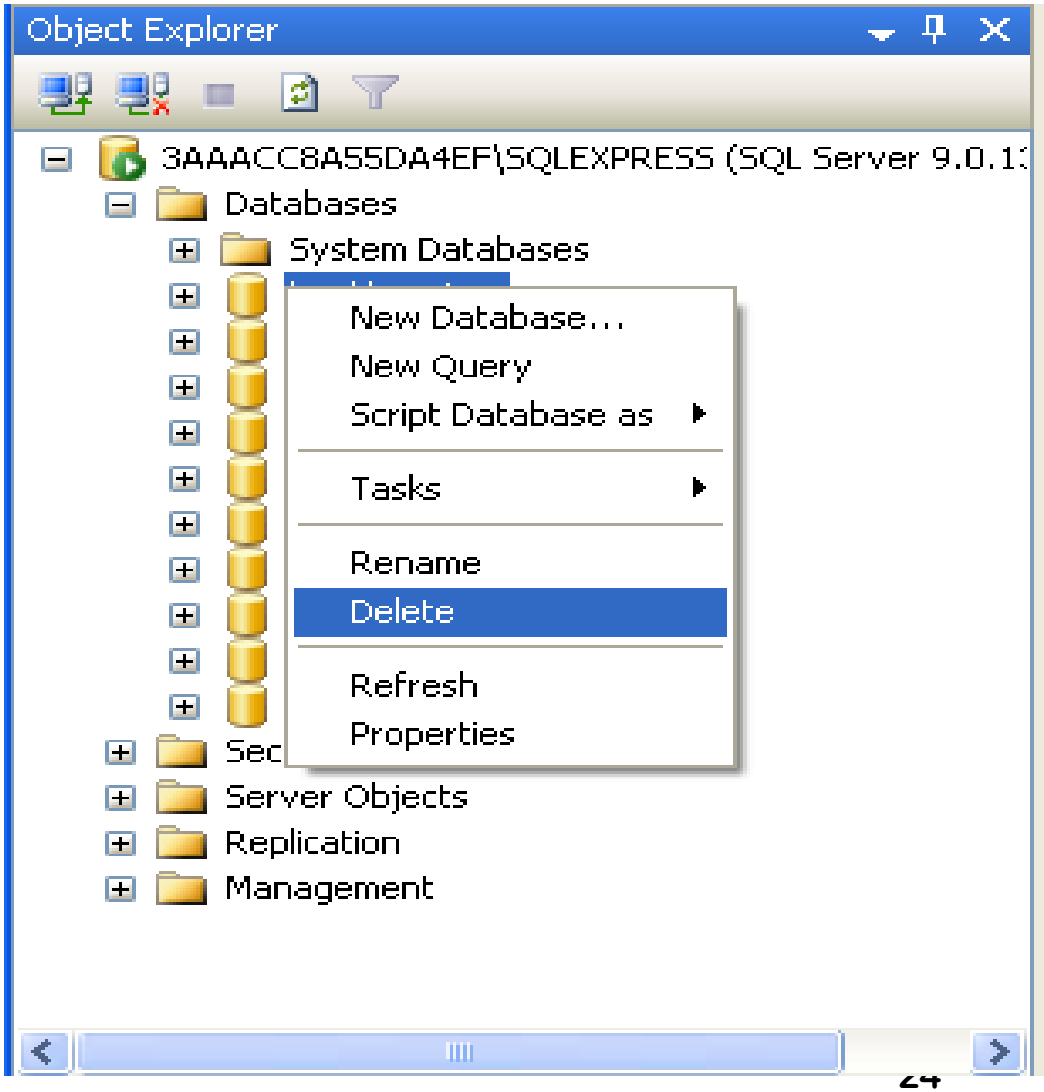
```
EXEC sp_renamedb  'accounting',
                  'financial'
```

Deleting a Database

Using Object Explorer

1. **Right-click on databases to be deleted**

2. **Select Delete**



Deleting a Database

Query Editor (T-SQL)

```
DROP DATABASE database_name [ ,...n ]
```

Deleting a Database

Query Editor (T-SQL)

```
DROP DATABASE publishing
```

```
DROP DATABASE pubs,newpubs
```

Creating Database

Using Object Explorer

1. Right-click on the Databases in the Object Explorer, then select New Databases…from the Context Menu that pops up, as you can see at the following figure:

2. Type **Database1** at the Database Name box.

3. Lastly, click the OK button

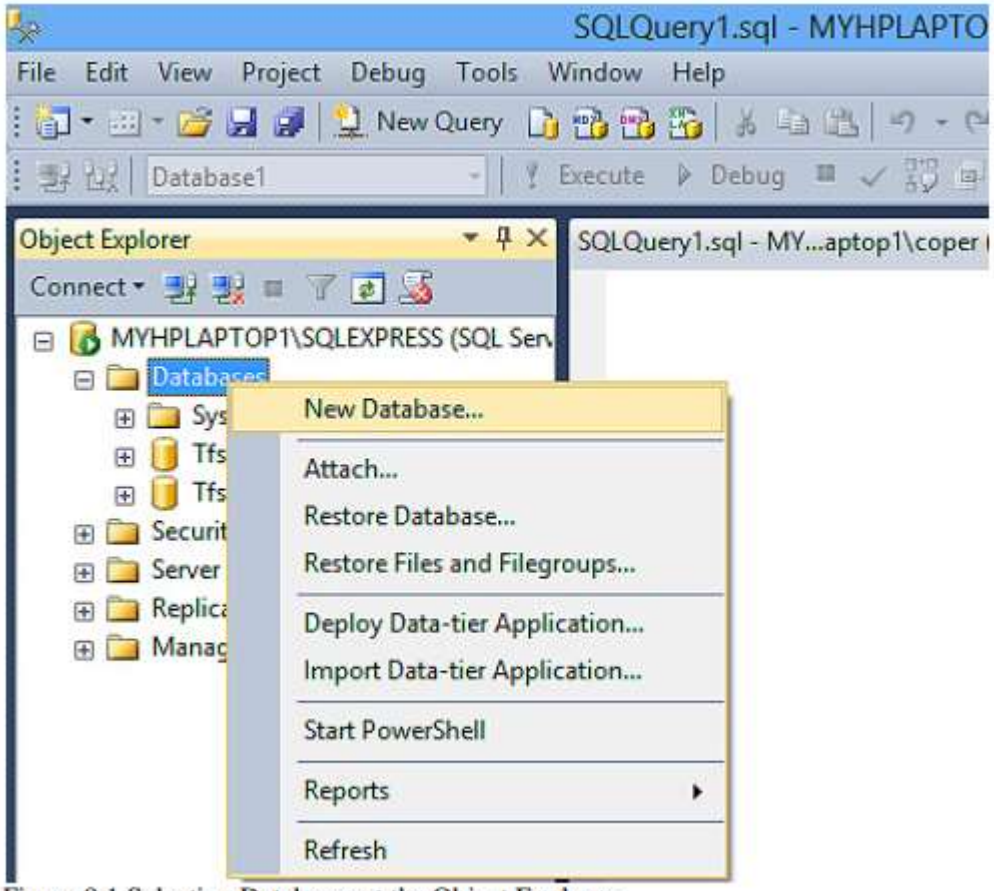


Figure 9.1 Selecting Databases at the Object Explorer

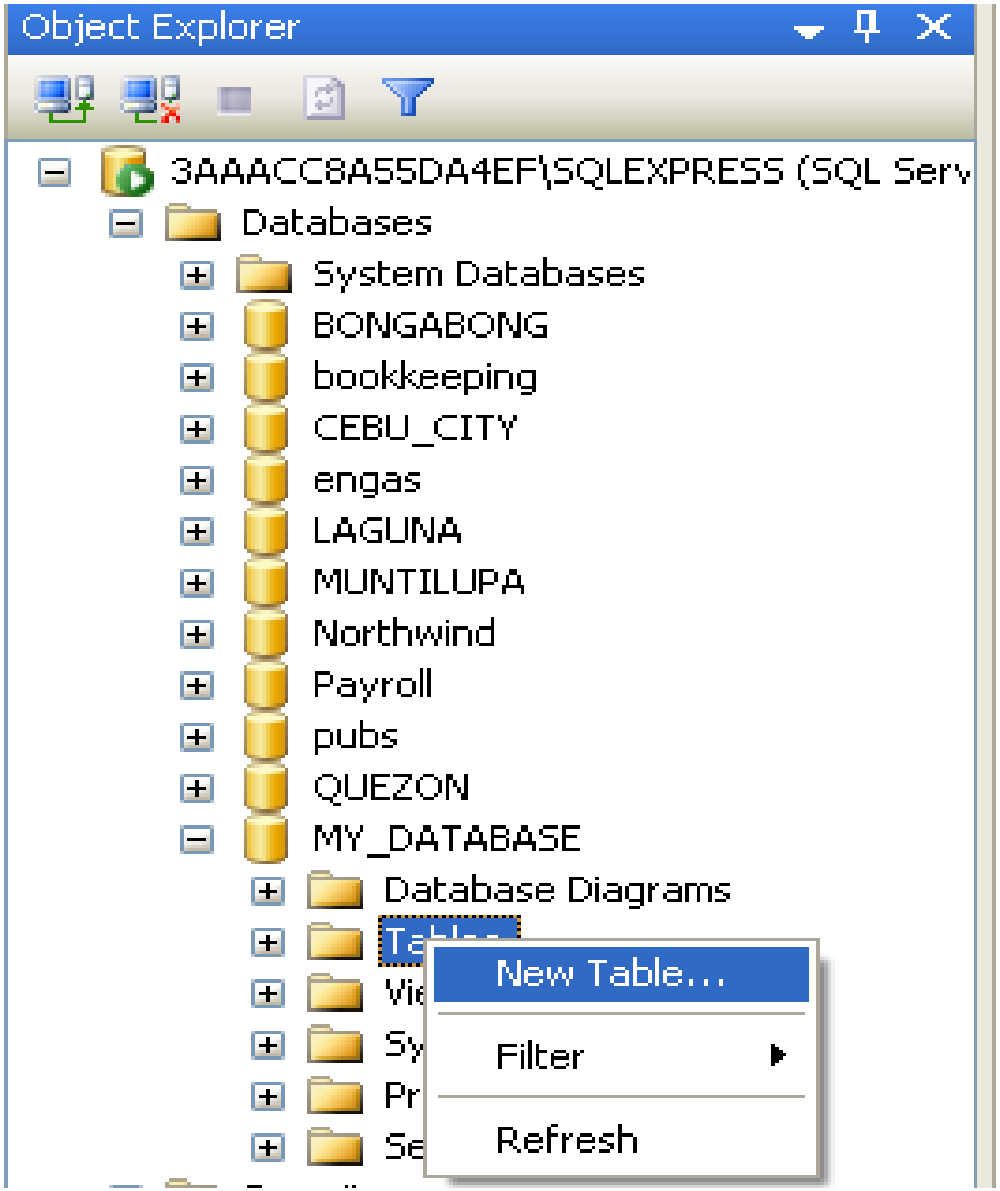
Creating Table

Using Object Explorer

1. Select and expand the Database where you want to create the table

2. Right-click Tables

3. Select New Table...

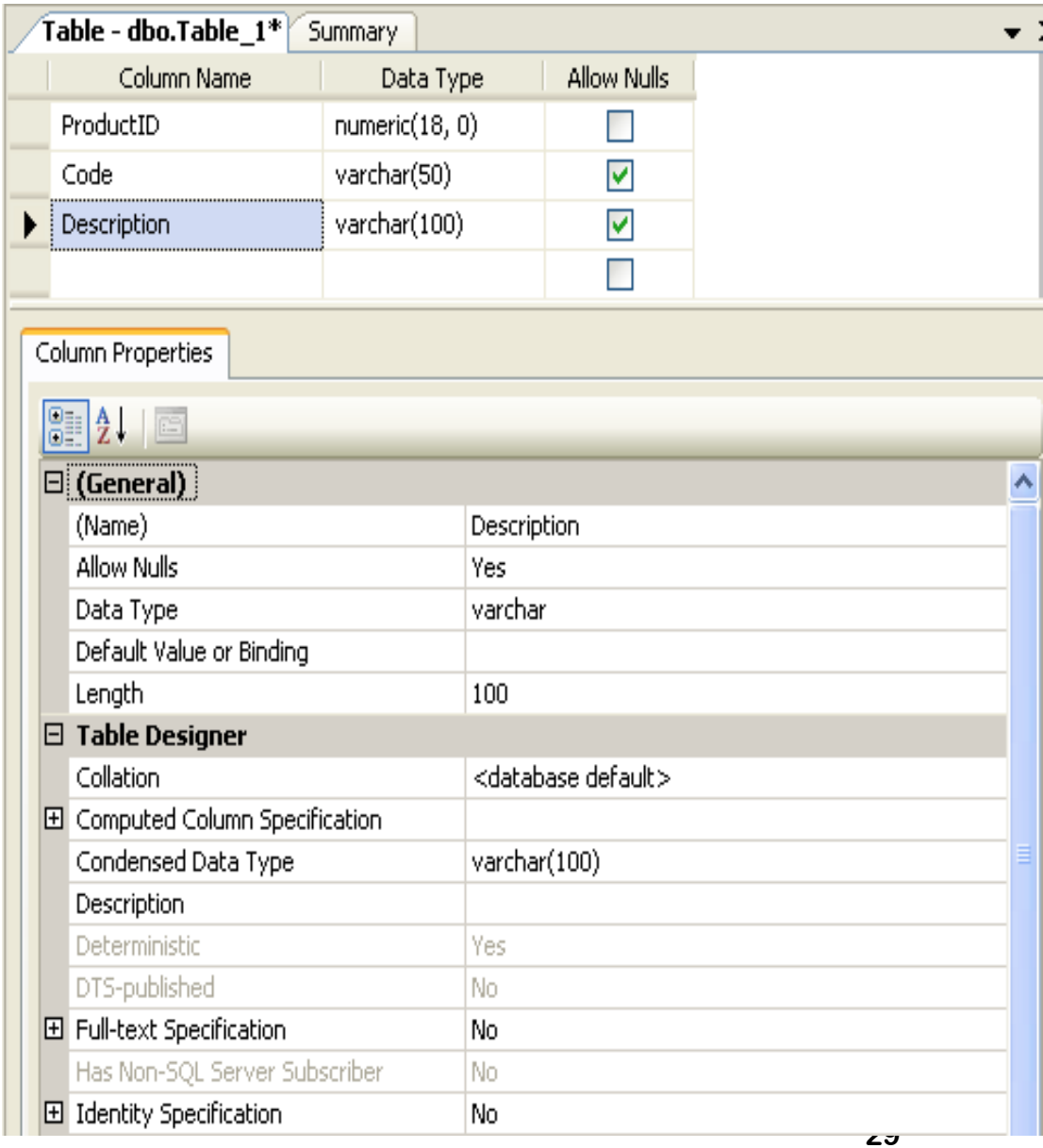


Creating Table

Using Object Explorer

4. In the **Table Designer**, supply the Column Name, Data Type and Allow Nulls

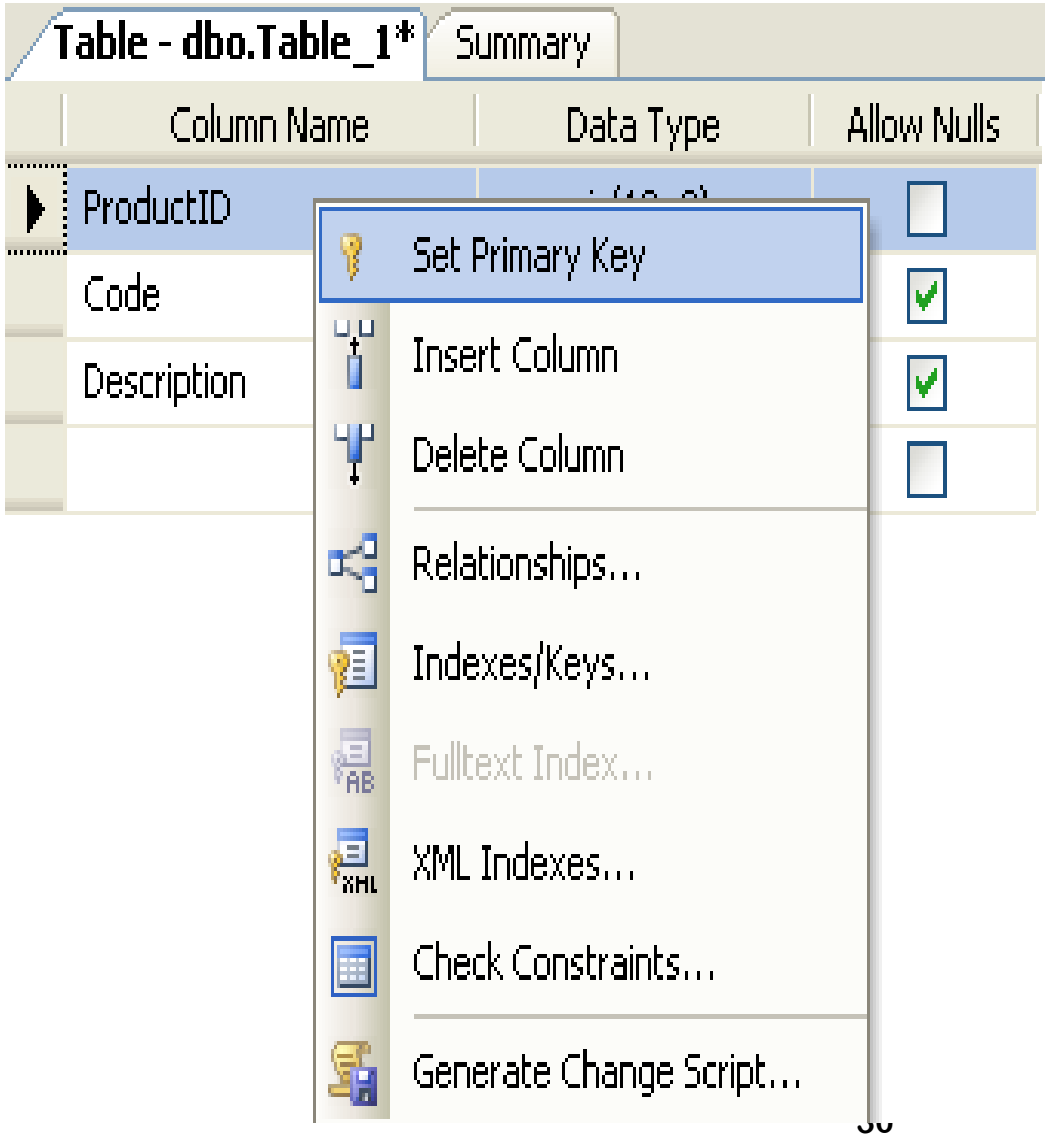
5. Below the column grid, you can modify the other properties of the column



Creating Table

Adding Primary Key

- Right-click column of interest
- Set Primary Key



Creating Table

Using Object Explorer

- Enter name of the table



Creating Table

Query Editor (T-SQL)

```
CREATE TABLE
  [ database_name. |owner. ]table_name
   ( column_name datatype
       [NULL | NOT NULL]
       [ DEFAULT constant_expression ]
       [ IDENTITY [(seed, increment)]
       [ , . . . n]
       [CONSTRAINTconstraint_name ]
  )
```

Creating Table

database_name

- **Optional. Name of the database in which the table will be created. Default is current database**

owner

- **Optional. Owner of the table. Default is *dbo***

table_name

- **Required. Name of the new table**

Creating Table

column_name

- Required. Name of the column

data_type

- Required. Data type of the column including the argument (length, precision, scale) if applicable

nullability

- Optional. Indicates whether null values are allowed in the column. Default is NULL

Creating Table

DEFAULT

- Optional. The value you want to be used for the column for any row that is inserted without explicitly supplying a value for that particular column

IDENTITY

- Optional. Indicates that the column is an identity column. The (seed, increment) values will default to (1,1) if not specified

Creating Table

```
CREATE TABLE dbo.Jobs
   (JobID numeric IDENTITY (1,1) NOT NULL
     CONSTRAINT PK_empID PRIMARY KEY,
    JobDescription VARCHAR(50) NOT NULL
   )
```

Creating Table

```
CREATE TABLE dbo.EMPLOYEES
  (empID numeric IDENTITY (1,1) NOT NULL
    CONSTRAINT PK_empID PRIMARY KEY,
    EmpNo varchar (20) NOT NULL,
    LName varchar (20) NOT NULL,
    FName varchar (20) NOT NULL,
    MName varchar (20),
    Active_Flag bit NOT NULL DEFAULT 1,
    JobID smallint NOT NULL
   CONSTRAINT FK_jobID FOREIGN KEY
   REFERENCES dbo.JOBS (JobID)
)
```

Viewing Information of a Table

Query Editor (T-SQL)

```
sp_help object_name
```

- The sp_help stored procedure is used to information about a database object in SQL Server

Viewing Information of a Table

Query Editor (T-SQL)

```
USE pubs
EXEC sp_help publishers
```

- This example displays information about the publishers table

Modifying Table

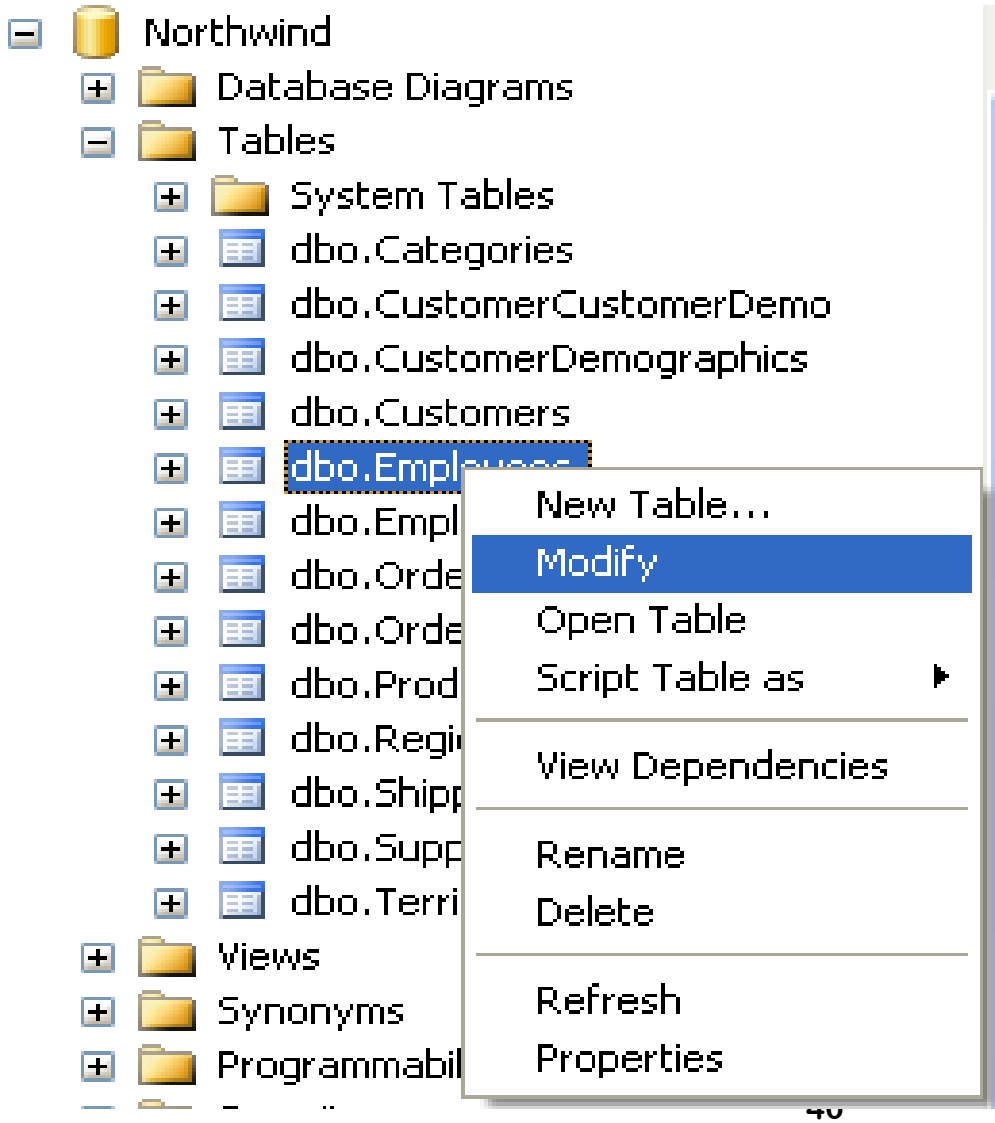
Using Object Explorer

1. Select and expand the Database where table exists

2. Expand Tables

3. Right click the table you want to edit

4. Select **Modify**



Modifying Table

Query Editor (T-SQL)

```
ALTER TABLE  table_name
 { [ ALTER COLUMN  column_name
     datatype} ]
     | ADD
        {[column definition]}
     | DROP
     {[CONSTRAINT]constraint_name
      |COLUMN column }
  }
```

Modifying Table

A. Alter a table to add a new column

```
CREATE TABLE doc_exa (column_a INT)
```

```
ALTER TABLE doc_exa
  ADD column_b VARCHAR(20) NULL
```

Modifying Table

B. Alter a table to drop a column

```
CREATE TABLE doc_exb (column_a INT,
    column_b VARCHAR(20) NULL)
```

```
ALTER TABLE doc_exb
  DROP column_b
```

Modifying Table

C. Alter a table to change the data type of a column

```
CREATE  TABLE  doc_exb (column_a INT,
      column_b VARCHAR(20) NULL)
```

```
ALTER  TABLE  doc_exb
  ALTER COLUMN column_b CHAR(20)
```

Modifying Table

D. Alter a table assign a constraint in an existing column

```
CREATE  TABLE  doc_exc (column_a INT,
      column_b VARCHAR(20) NULL)
```

```
ALTER  TABLE  doc_exc
  ADD CONSTRAINT [PK_doc_exc ]
    PRIMARY KEY (column_a)
```

Modifying Table

E. Renaming table name

```
sp_rename 'doc_exb',
          'doc_ex_b'
```

Modifying Table

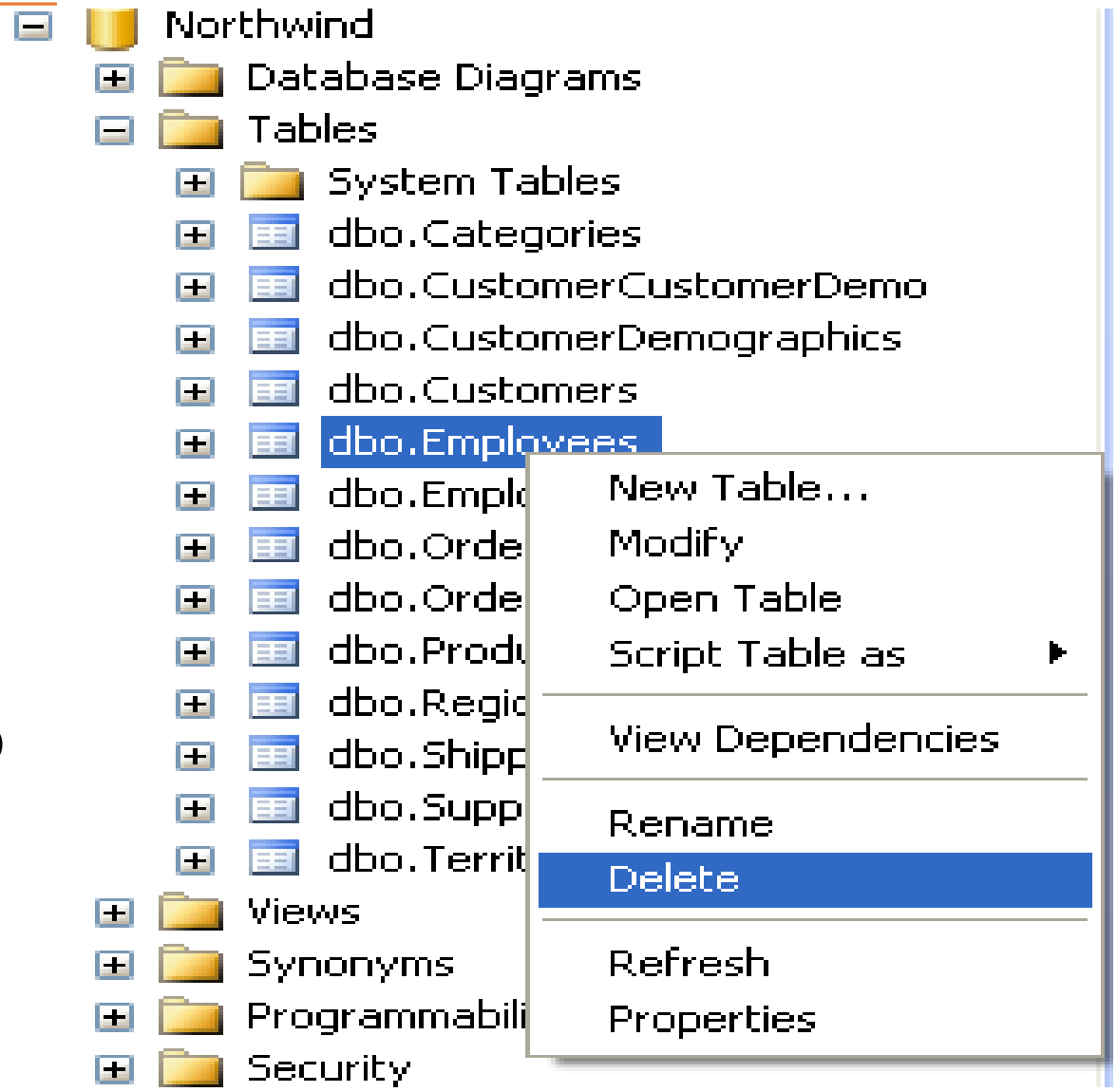
F. Renaming column name of a table

```
sp_rename 'doc_ex_b.[column_a]',
          'col_a', 'column'
```

Dropping Table

Using Object Explorer

1. Select and expand the Database where table exists
2. Expand Tables
3. Right click the table you want to delete
4. Select **Delete**



Dropping Table

Query Editor (T-SQL)

```
DROP TABLE table_name
```

Dropping Table

Query Editor (T-SQL)

```
DROP TABLE doc_exa
```

```
DROP TABLE doc_ex_b, doc_exc
```